

MADHU CHITIKELA

AI/ML Engineer • NLP • AWS

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PROFESSIONAL SUMMARY

Final-year B.Tech AI/ML student and Oracle-certified AI Professional with hands-on experience building and deploying AI systems using LangChain, BERT, FastAPI, Docker, and AWS. Developed LLM-powered agents, NLP pipelines, and CI/CD-enabled ML deployment workflows with practical exposure to scalable application design and cloud infrastructure. Led teams in ABB Accelerator 2026 and Cognizant Technoverse 2026 hackathons. Seeking a GenAI or ML engineering internship to contribute strong problem-solving and deployment-focused engineering skills in a high-impact environment.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

AI / ML: Machine Learning, Deep Learning, NLP, Generative AI, Model Fine-tuning, SHAP Explainability, IRT Algorithm

LLM & Agents: LangChain, Groq API, Gemini API, Tavily API, HuggingFace Transformers, BERT, ReAct Agent Pattern

Frameworks: FastAPI, Streamlit, Gradio, TensorFlow, PyTorch, Scikit-learn, Next.js, React, TailwindCSS, Pydantic, pytest

Cloud & DevOps: AWS EC2, S3, IAM, Lambda, Docker, docker-compose, GitHub Actions (CI/CD), Git, Oracle Cloud Infrastructure, Vercel

Data Tools: Pandas, NumPy, Matplotlib, Plotly, Jupyter Notebook

Databases: SQLite, MySQL

CS Fundamentals: DSA, OOP, DBMS, Operating Systems, Computer Networks, Linux

PROJECTS

LLM-Powered Code Debugger Agent [GitHub] [Live Demo]

- Designed a LangChain ReAct agent that debugs broken Python code via iterative tool-calling loops; modular architecture separates agent logic, tool definitions, and UI.
- Built a 7-model fallback system across Groq (LLaMA 3.3, Gemma2, Mixtral) and Gemini, reducing failures during API rate-limit events.
- Added SQLite-backed session logging and a real-time trace dashboard in Gradio; validated against 30+ broken code samples across different error categories.

Stack: Python · LangChain · Gradio · HuggingFace Spaces

ML Model Deployment Pipeline — CI/CD on AWS [GitHub]

- Built a GitHub Actions CI/CD workflow that builds, runs pytest unit tests, lints with flake8, and deploys a containerized FastAPI ML model to AWS EC2 on every push to main.
- Configured AWS S3 for model artifact storage and IAM roles for least-privilege access; automated test checks gate every deployment.
- Validated the full Docker → ECR → EC2 deployment workflow end-to-end with a health-check endpoint confirming live inference.

Stack: Python · FastAPI · Docker · AWS EC2 · GitHub Actions

Fake News Detector — AI News Verification [GitHub] [Live Demo]

- Fine-tuned BERT (bert-base-uncased) on 45,000+ news articles using NVIDIA RTX 2050 GPU; evaluated on a held-out 20% test split to validate generalization performance.
- Engineered a dual-layer pipeline: BERT classifies writing patterns; a LangChain + Tavily agent verifies factual claims against live sources in real time.
- Integrated SHAP explainability with tiered risk scores (HIGH / MEDIUM / LOW); deployed 438 MB model via Git LFS supporting concurrent Streamlit sessions.

Stack: Python · BERT · HuggingFace Transformers · Streamlit

Adaptive Learning Coach — Personalized AI Tutor [GitHub] [Live Demo]

- Implemented a 3-Parameter psychometric engine based on Item Response Theory (IRT) modeling to estimate student ability in real time and select optimal-difficulty questions.
- Integrated a memory-enabled LangChain coach that retains conversation history and personalizes explanations based on diagnosed weak areas.
- Auto-generates 7-day study plans from diagnostic results; dashboard renders learning curves, radar charts, and score gauges.

Stack: Python · LangChain · Groq · IRT Algorithm · HuggingFace Spaces

HACKATHONS & LEADERSHIP

ABB Accelerator 2026 — Team Leader

2026

- Led a 4-member team in an engineering innovation competition focused on industrial AI problem-solving; coordinated project architecture, task distribution, and technical decision-making.

Cognizant Technoverse 2026 · ResilienceIQ — Team Leader

2026

- Built an AI/ML climate risk and catastrophe insurance platform using SHAP/LIME explainability. Competed against 500+ teams; advanced to the final evaluation round.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 — Generative AI Professional

Oracle · 2025

Oracle Cloud Infrastructure 2025 — AI Foundations Associate

Oracle · 2025

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning — CGPA: 8.96 / 10

2023 – 2027

GIET Engineering College, Rajamahendravaram, Andhra Pradesh

Intermediate (MPC) — 93%

2021 – 2023

Narayana Junior College

Matriculation (SSC) — CGPA: 9.93 / 10

2021

Sri Chaitanya E.M. School